

DSA Pattern Sheet

1. Arrays & Strings

1.1 Two Pointers

What it is:

Use two indices moving towards each other or in the same direction.

Used for:

Pairs, comparisons, partitioning, removing duplicates

LeetCode Problems:

- <https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/>
 - <https://leetcode.com/problems/valid-palindrome/>
 - <https://leetcode.com/problems/remove-duplicates-from-sorted-array/>
 - <https://leetcode.com/problems/container-with-most-water/>
 - <https://leetcode.com/problems/squares-of-a-sorted-array/>
 - <https://leetcode.com/problems/trapping-rain-water/>
 - <https://leetcode.com/problems/3sum/>
 - <https://leetcode.com/problems/4sum/>
-

1.2 Sliding Window (Fixed & Variable)

What it is:

Maintain a window and adjust it based on constraints.

Used for:

Subarrays, substrings, max/min, counts

LeetCode Problems:

- <https://leetcode.com/problems/maximum-sum-of-subarray-of-size-k/>
- <https://leetcode.com/problems/longest-substring-without-repeating-characters/>
- <https://leetcode.com/problems/minimum-size-subarray-sum/>
- <https://leetcode.com/problems/longest-repeating-character-replacement/>
- <https://leetcode.com/problems/permutation-in-string/>
- <https://leetcode.com/problems/find-all-anagrams-in-a-string/>
- <https://leetcode.com/problems/fruit-into-baskets/>

1.3 Prefix Sum

Used for:

Range queries, subarray sum problems

LeetCode Problems:

- <https://leetcode.com/problems/subarray-sum-equals-k/>
 - <https://leetcode.com/problems/range-sum-query-immutable/>
 - <https://leetcode.com/problems/find-pivot-index/>
 - <https://leetcode.com/problems/continuous-subarray-sum/>
-

1.4 Kadane's Algorithm

Used for:

Maximum subarray type problems

LeetCode Problems:

- <https://leetcode.com/problems/maximum-subarray/>
 - <https://leetcode.com/problems/maximum-product-subarray/>
-

1.5 Cyclic Sort

Used for:

Missing / duplicate numbers

LeetCode Problems:

- <https://leetcode.com/problems/missing-number/>
 - <https://leetcode.com/problems/find-all-numbers-disappeared-in-an-array/>
 - <https://leetcode.com/problems/find-the-duplicate-number/>
 - <https://leetcode.com/problems/first-missing-positive/>
-

1.2 Sliding Window (Fixed & Variable)

What it is:

Maintain a window range and slide it while keeping constraints.

Used for:

- Subarray / substring
- Max / min / longest / count

Problems:

- <https://leetcode.com/problems/maximum-sum-of-subarray-of-size-k/>
 - <https://leetcode.com/problems/longest-substring-without-repeating-characters/>
 - <https://leetcode.com/problems/minimum-size-subarray-sum/>
-

1.3 Prefix Sum

What it is:

Store cumulative sum to answer range queries efficiently.

Used for:

- Subarray sum
- Range queries

Problems:

- <https://leetcode.com/problems/subarray-sum-equals-k/>
 - <https://leetcode.com/problems/range-sum-query-immutable/>
-

1.4 Kadane's Algorithm

What it is:

Optimized DP for max subarray problems.

Problems:

- <https://leetcode.com/problems/maximum-subarray/>
-

1.5 Cyclic Sort

What it is:

Place numbers at correct indices.

Used for:

- Missing numbers
- Duplicates

Problems:

- <https://leetcode.com/problems/missing-number/>
 - <https://leetcode.com/problems/find-all-numbers-disappeared-in-an-array/>
-

2. Hashing

2.1 Frequency Map

Used for:

Counting, existence, complements

LeetCode Problems:

- <https://leetcode.com/problems/two-sum/>
 - <https://leetcode.com/problems/valid-anagram/>
 - <https://leetcode.com/problems/group-anagrams/>
 - <https://leetcode.com/problems/top-k-frequent-elements/>
 - <https://leetcode.com/problems/first-unique-character-in-a-string/>
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2.2 Hash Set Pattern

Used for:

Detect duplicates, cycles

LeetCode Problems:

- <https://leetcode.com/problems/contains-duplicate/>
 - <https://leetcode.com/problems/longest-consecutive-sequence/>
-

3. Linked List

3.1 Fast & Slow Pointer

LeetCode Problems:

- <https://leetcode.com/problems/linked-list-cycle/>
 - <https://leetcode.com/problems/middle-of-the-linked-list/>
 - <https://leetcode.com/problems/palindrome-linked-list/>
-

3.2 Reversal Pattern

LeetCode Problems:

- <https://leetcode.com/problems/reverse-linked-list/>
 - <https://leetcode.com/problems/reverse-nodes-in-k-group/>
 - <https://leetcode.com/problems/swap-nodes-in-pairs/>
-

3.3 Merge Pattern

LeetCode Problems:

- <https://leetcode.com/problems/merge-two-sorted-lists/>
 - <https://leetcode.com/problems/merge-k-sorted-lists/>
-

3.2 Reversal Pattern

Used for:

- Reverse entire list or parts

Problems:

- <https://leetcode.com/problems/reverse-linked-list/>
 - <https://leetcode.com/problems/reverse-nodes-in-k-group/>
-

4. Stack & Queue

4.1 Monotonic Stack

LeetCode Problems:

- <https://leetcode.com/problems/next-greater-element-i/>

- <https://leetcode.com/problems/daily-temperatures/>
 - <https://leetcode.com/problems/largest-rectangle-in-histogram/>
 - <https://leetcode.com/problems/trapping-rain-water/>
-

4.2 Queue / Deque Pattern

LeetCode Problems:

- <https://leetcode.com/problems/sliding-window-maximum/>
 - <https://leetcode.com/problems/design-circular-deque/>
-

4.2 Queue / Deque Pattern

Used for:

- Sliding window max/min

Problems:

- <https://leetcode.com/problems/sliding-window-maximum/>
-

5. Binary Search

5.1 Classic Binary Search

Problems:

- <https://leetcode.com/problems/binary-search/>
-

5.2 Binary Search on Answer

Used for:

- Min/max optimization

Problems:

- <https://leetcode.com/problems/koko-eating-bananas/>

- <https://leetcode.com/problems/split-array-largest-sum/>
-

6. Recursion & Backtracking

6.1 Pick / Not Pick

Used for:

- Subsets
- Combinations

Problems:

- <https://leetcode.com/problems/subsets/>
 - <https://leetcode.com/problems/combinations/>
-

6.2 Backtracking

Problems:

- <https://leetcode.com/problems/permutations/>
 - <https://leetcode.com/problems/n-queens/>
-

7. Trees

7.1 DFS (Pre/In/Post)

Problems:

- <https://leetcode.com/problems/binary-tree-inorder-traversal/>
 - <https://leetcode.com/problems/maximum-depth-of-binary-tree/>
-

7.2 BFS (Level Order)

Problems:

- <https://leetcode.com/problems/binary-tree-level-order-traversal/>
-

8. Graphs

8.1 BFS & DFS

LeetCode Problems:

- <https://leetcode.com/problems/number-of-islands/>
 - <https://leetcode.com/problems/flood-fill/>
 - <https://leetcode.com/problems/max-area-of-island/>
 - <https://leetcode.com/problems/rotting-oranges/>
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8.2 Topological Sort

LeetCode Problems:

- <https://leetcode.com/problems/course-schedule/>
 - <https://leetcode.com/problems/course-schedule-ii/>
-

8.3 Union Find

What it is:

Disjoint Set data structure to manage dynamic connectivity.

Used for:

- Detecting cycles
- Connected components

LeetCode Problems:

- <https://leetcode.com/problems/redundant-connection/>
 - <https://leetcode.com/problems/number-of-provinces/>
 - <https://leetcode.com/problems/accounts-merge/>
-

8.4 Dijkstra's Algorithm (Shortest Path)

What it is:

Greedy algorithm using a min-heap to find the shortest path from a source to all nodes (non-negative weights).

Used for:

- Shortest path problems
- Weighted graphs
- Minimum cost / time problems

LeetCode Problems:

- <https://leetcode.com/problems/network-delay-time/>
 - <https://leetcode.com/problems/path-with-minimum-effort/>
 - <https://leetcode.com/problems/cheapest-flights-within-k-stops/>
 - <https://leetcode.com/problems/minimum-cost-to-reach-destination-in-time/>
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8.2 Topological Sort

Problems:

- <https://leetcode.com/problems/course-schedule/>
-

8.3 Union Find

Problems:

- <https://leetcode.com/problems/redundant-connection/>
-

9. Heap

9.1 Top K Pattern

Problems:

- <https://leetcode.com/problems/kth-largest-element-in-an-array/>
 - <https://leetcode.com/problems/top-k-frequent-elements/>
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10. Dynamic Programming

10.1 0/1 Knapsack

LeetCode Problems:

- <https://leetcode.com/problems/partition-equal-subset-sum/>
 - <https://leetcode.com/problems/target-sum/>
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10.2 Unbounded Knapsack

LeetCode Problems:

- <https://leetcode.com/problems/coin-change/>
 - <https://leetcode.com/problems/coin-change-2/>
-

10.3 DP on Strings

LeetCode Problems:

- <https://leetcode.com/problems/longest-common-subsequence/>
 - <https://leetcode.com/problems/edit-distance/>
 - <https://leetcode.com/problems/longest-palindromic-subsequence/>
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10.2 Unbounded Knapsack

Problems:

- <https://leetcode.com/problems/coin-change/>
-

10.3 DP on Strings

Problems:

- <https://leetcode.com/problems/longest-common-subsequence/>
 - <https://leetcode.com/problems/edit-distance/>
-

11. Greedy

What it is:

Make the locally optimal choice at each step, trusting it leads to a global optimum.

Used for:

Intervals, scheduling, optimization, resource allocation

11.1 Interval & Scheduling Greedy

LeetCode Problems:

- <https://leetcode.com/problems/non-overlapping-intervals/>
 - <https://leetcode.com/problems/merge-intervals/>
 - <https://leetcode.com/problems/meeting-rooms/>
 - <https://leetcode.com/problems/meeting-rooms-ii/>
 - <https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/>
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11.2 Jump / Reachability Greedy

LeetCode Problems:

- <https://leetcode.com/problems/jump-game/>
 - <https://leetcode.com/problems/jump-game-ii/>
 - <https://leetcode.com/problems/gas-station/>
-

11.3 Optimization Greedy

LeetCode Problems:

- <https://leetcode.com/problems/assign-cookies/>
 - <https://leetcode.com/problems/boat-to-save-people/>
 - <https://leetcode.com/problems/task-scheduler/>
 - <https://leetcode.com/problems/reorganize-string/>
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12. Bit Manipulation

What it is:

Use bitwise operators to solve problems efficiently.

Used for:

XOR tricks, subsets, power of two checks, bit masks

12.1 XOR Based Problems

LeetCode Problems:

- <https://leetcode.com/problems/single-number/>
 - <https://leetcode.com/problems/single-number-ii/>
 - <https://leetcode.com/problems/single-number-iii/>
-

12.2 Bit Masking

LeetCode Problems:

- <https://leetcode.com/problems/subsets/>
 - <https://leetcode.com/problems/subsets-ii/>
 - <https://leetcode.com/problems/letter-case-permutation/>
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12.3 Bit Tricks & Counting

LeetCode Problems:

- <https://leetcode.com/problems/counting-bits/>
 - <https://leetcode.com/problems/number-of-1-bits/>
 - <https://leetcode.com/problems/reverse-bits/>
 - <https://leetcode.com/problems/power-of-two/>
 - <https://leetcode.com/problems/missing-number/>
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12.4 Advanced Bit Manipulation

LeetCode Problems:

- <https://leetcode.com/problems/maximum-xor-of-two-numbers-in-an-array/>
 - <https://leetcode.com/problems/bitwise-and-of-numbers-range/>
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